

ACM MIGRATION AZURE – Non-SAP LOW LEVEL DESIGN

(LLD001A)

Microsoft Azure

accenture

Pipeline Pilot – R&D, Global

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1.1	25-Apr-2022	Harmohan Singh		Removed App Proxy from Solution. Added ZIA in design diagram, for Internet Access.

GOVERNANCE & APPROVAL		
Reviewed By	**Role**	**Date**
Nitika Gupta	Design Peer Reviewer (mandatory)	**1-Mar-22**
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1. OVERVIEW

The following sub-sections provide the reader with the business context and scope for this solution.

1.1 Business Description

Pipeline Pilot (PLP) is a key tool from Accelrys (now owned by Biovia) for the informatics, AST, and new Biology work streams. It is a key data and informatics system for Unilever R & D (specifically eScience program) and is a middleware application that facilitates coordination of workflows in several scientific applications, few examples being R&D ELN and EMS.

1.2 Scope

The scope of this migration is single sandbox server which is currently hosted in UK DC. All other environments (Prod, Stage, QA & Dev) are already hosted in Azure Amsterdam region.

- App Server cgtsapp24463 will be migrated from On-premises to Azure IaaS Virtual Machines in West Europe.
- The cgtsapp24463 server is running on Windows 2016 standard edition and it will be upgraded to Windows 2019 Datacenter edition in Cloud.
- Migration methodology for this server will be side-by-side migration approach wherein new server will be built in cloud and the application team will perform the installation & configuration of the application with the help of the vendor.
- The current application version is 2019 and it will be upgraded to 2020 version as part of the migration, by the application team.
- Azure Blob will be provisioned which will be used for migrating business & non-business data from On-Prem server to Azure and for archiving the protocols post migration.
- Existing application server has shared Drive (\\CGTSAPP24463\Public) which the users are required to mount on their local system, and it will be migrated to Azure File Share (AFS).

1.3 Application Context Diagram

Not Available.

2. SERVICE, RISK AND EXCEPTIONS

The following section provides the reader with a view of the service and data criticality including any architecture standards deviations, security exceptions and risks associated with the design.

2.1 Information Criticality and Availability Assessment (ICAA)

The following section provides the reader with an understanding of the service and data criticality details for the design. Note. The ITSG and ISA details can be provided by the project.

Environment	ICAA Ref	Data Classification	SC Class	% Uptime	DR & RTO	RPO
Sandbox*	CMDB	Confidential	SC4	97	DR4 (≤ 2 weeks)	7 days

Note. As part of the Secure by Design (SBD) process, a Self-Assessment (SA) or Information Security Assessment (ISA) is completed by the Service Owner if the data classification is rated as Internal. Where the data classification is rated as confidential or above a dedicated Information Security Analyst is assigned to complete the Information Security Assessment (ISA). Public data may be assessed differently.

* ICCA for Prod is 7611.

2.2 Risks, Architecture & Security Exceptions

The following table provides the reader with an understanding any architecture and / or security exceptions required for the design. Note. Project implementation and build RAID items should be captured in the project RAID log.

Category	Description
Architecture Standards Deviation and Security Exceptions	ZScalar Internet Access (ZIA) will be used to provide Internet access on the Azure VM, because IT Platform would like controlled internet access without URL whitelisting. Although, this is an architectural standard deviation, it does not conflict with security policy.
Risks	There is a risk for Azure cloud hosted services with a DR classification of 3 and below where dedicated capacity in the DR paired region has not been reserved that in the event of a regional disaster, capacity from the vendor may not be available for Unilever resources. Same has been communicated with the platform team.

3. KEY DESIGN DECISIONS (KDD)

The following table provides the reader with a view of the Key Design Decisions taken for the Low Level Design.

Ref.	Key Design Decision Area	Key Design Decision	Justification
1	Resilience (HA)	HA will not be deployed.	Aligns to SC principle. High availability is not required for SC4 classified applications.
2	DR	No DR will be deployed.	Aligns to DR principle. Sandbox environment does not require a DR to be built.
3	Hosting Location	Application Server will be hosted in Azure Amsterdam Region (West Europe) Subscription: R&D-In Silico First eScience-01	Application server will be migrated from on-premises DC to Azure subscription as a part of ACM-Non-SAP migration.
4	Hosting Type	VMs for Application to be hosted as IaaS.	Side-by-side migration approach from on-premises to Azure IaaS.
5	SBBs / Modules	The design is using SBBs and modules only.	The design is using standard infrastructure and design diagram templates.
6	Security	Some connections are not secured.	As per security standards most of the connections will be encrypted for data in transit. There is an HTTP port open on the server, which is used to facilitate tomcat management, wherein, connection is initiated from the server loopback address – locally.
7	Security	Citrix Gateway will be used.	As per security standards, the Citrix interface will be used for access to Cloud Servers.

Ref.	Key Design Decision Area	Key Design Decision	Justification
8	Security	Application will be accessible over the internal network only.	Currently, the application is not published to internet. The application will remain internal to Unilever Network and will be accessed via Express Route & CXP firewall when working in office and ZPA when working from home.
9	Security	ZScalar Internet Access will be used.	IT Platform would like controlled internet access without URL whitelisting. See architecture & deviation section 2.2 for more details. NAT Gateway will be used in order to register the IP address in the ZScalar.
10	Integration	Integration with other systems.	This server is integrated with rpli001a Linux server running in PSC building (Port Sunlight).

Please see **Section 2.2** for the architecture and security exceptions required that relate to the above KDDs (if applicable).

4. DEPLOYMENT

The following section provides the reader with a view of the landscape components including the hosting location, infrastructure and applications installed on them. It also defines the High Availability and Disaster Recovery architecture for the solution.

4.1.1 Production

The production environment is hosted in Azure, and it consists of a Windows VM IaaS instance with Apache web server and Pipeline Pilot Scientific Platform 2018 installed, connecting to a MS SQL server 2016 database (IaaS Instance) in the backend. Please refer to the existing SDL for more details. Link provided in appendices.

Production is out-of-scope of this migration.

4.1.2 Service High Availability (HA) Architecture

Not applicable.

4.1.3 Service Disaster Recovery (DR) Architecture or Process

Not applicable.

4.1.4 Non-Production

- The existing Pipeline Pilot - R&D, Global Non-Prod service has 5 application servers, three of which are already provisioned in Azure. Other two servers are cgtsapp24463 and rpli001a.
- Server cgtsapp24463 is hosted in UK DC and will be migrated from On-premises to Azure IaaS Virtual Machines in West Europe.
- Server rpli001a is part of Port Sunlight Crossroads project and it is situated in the PSC building. This will be decommissioned by end of 2022.
- Below ports will required to be opened on cgtsapp24464 app server to facilitate technical management of tomcat (Note – the management tasks are initiated from the server loopback address – locally. There is no data moving on to this connection.) –
    - 947 (Tomcat shutdown)
    - 9946 (Tomcat HTTP)
    - 9947 (Derby Port)
- cgtsapp24463 consist of two network drives which can be accessed by users on their local systems and these drives will be replaced with Azure File shares.
- Storage BLOB will be provisioned as requested by the platform team to use it for archiving application protocols.
- An SFTP Connection is initiated from cgtsapp24463 by the application. It's an inbuilt function in the app to fetch data/files from a location internally. No explicit SFTP client is required to be installed on the server.

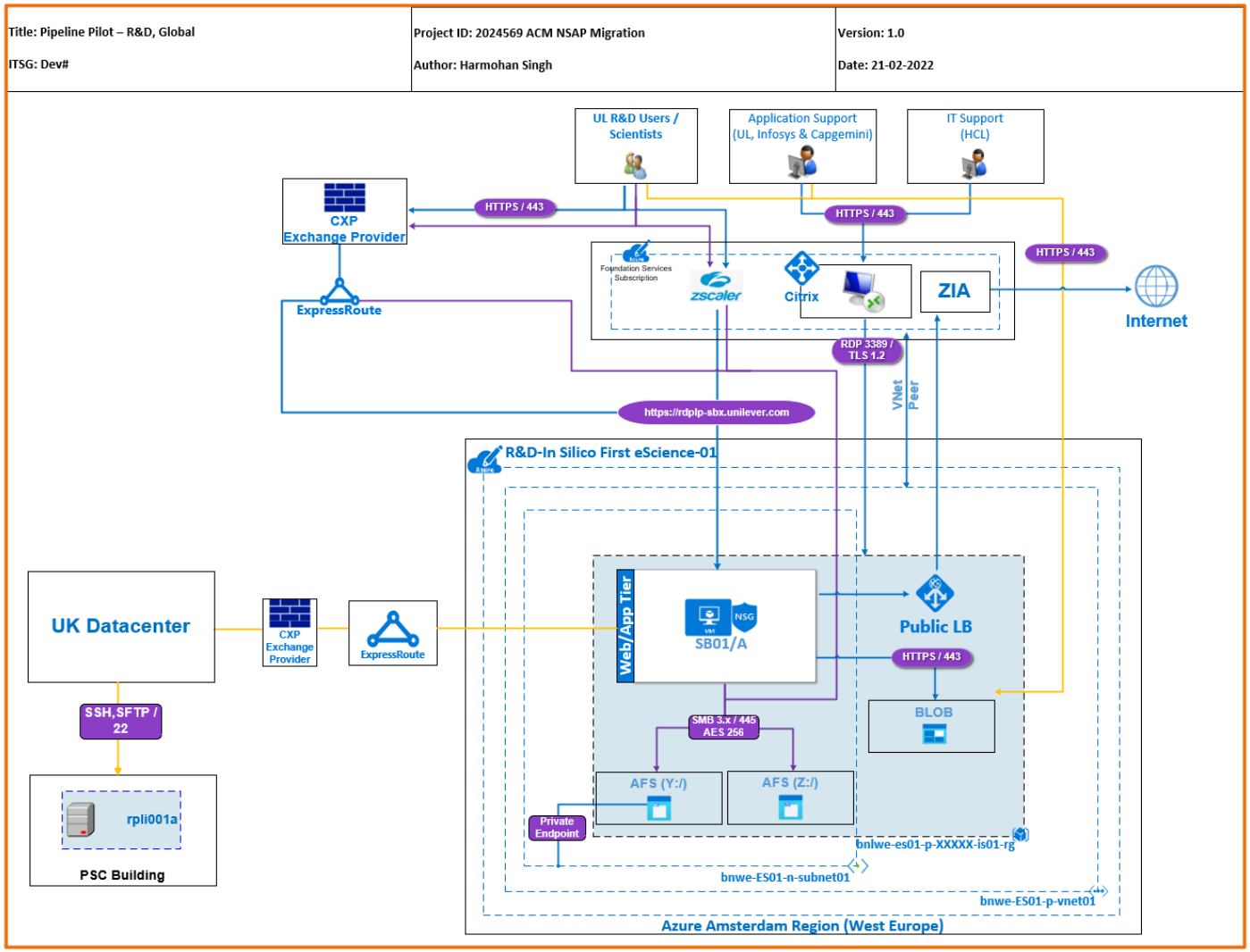
4.2 Landscape Diagram

4.2.1 Production

NA

4.2.2 Non-Production

Title: Pipeline Pilot – R&D, Global	Project ID: 2024569 ACM NSAP Migration	Version: 1.0
ITSG: Dev#	Author: Harmohan Singh	Date: 21-02-2022



5. LICENSING, ASSUMPTIONS AND DEPENDENCIES

5.1 Assumptions, Issues & Dependencies

The following table provides the reader with an understanding of the **architecture items** only. Note. Project implementation and build RAID items should be captured in the project RAID log.

Category	Description
Assumptions	None
Issues	None
Dependencies	There is a requirement to enable Internet access on the application Server without URL whitelisting. ZScalar Internet Access (ZIA) will be used to provide Internet access on the Azure VM.

5.2 Service Life and Decommissioning

The following table provides the reader with an understanding of the solutions service life and any decommissioning required.

Category	Description
Service Life	See service owner.
Decommissioning	The On-Prem Server cgtsapp24463 will be decommissioned after migration to Azure cloud and post 2 weeks of PGLS. The on-premises infra decommissioning should be initiated by the platform team.
IT System Group (ITSG)	TBD

5.3 Application Software Components

The following table provides the reader with the major application software components and versions to be implemented as part of this solution.

Software Component Name	Software Component Version
Pipeline Pilot Software	2020

5.4 Licences

The following table provides the reader with the licenses required for the infrastructure deployment. The table below specifies the technical amount required in and if the licenses come in packs or groups this is detailed in the subsection below (Additional Licensing Notes).

Software Licence	Subscription Qty	Site Qty	CPU Socket Qty	CPU Core Qty	Server Qty	Client Qty	BYOL or Dual-Use
Windows Server 2019 DC Edition	**1**	**-**	**-**	**-**	**-**	**-**	**-**

5.4.1 Additional Licensing Notes

None

6. INTEGRATION ARCHITECTURE

6.1 Interfaces List

The following table provides the reader with a view of the technical integration for the deployment. Note. Infrastructure tooling and foundation services interfaces do not need to be added to this section as these are part of the Foundation Services Standard Building Blocks (SBBs).

Ref	Source	Middleware or Infrastructure Component Facilitating Data Flow	Target	Batch (B) / Transactional (T)	Protocol / Port	Encrypted
1	Citrix	Citrix Gateway	App Server	T	RDP / 3389 TLS 1.2	Yes
2	UL Internal Network (Laptops / Desktop)	Express Route, ZPA	App Server	T	HTTPS / 443	Yes
3	UL R&D Users	-	Storage Blob	T	HTTPS / 443	Yes
3	UL Laptops / Desktop	ZPA, Express Route	Azure File Share	T	SMB 3.x / 445 AES 256	Yes
4	App Server	Express Route	Rpli001a	T	SSH, SFTP / 22	Yes
5	App Server	NSG	Storage blob	T	HTTPS / 443	Yes
6	App Server	NSG	Azure File Share	T	SMB 3.x / 445 AES 256	Yes
7	App Server	ZScalar Internet Access	Internet (SaaS Services)	-	-	-

6.2 Cloud to / from On-Prem Interfaces – Extra Detail (If Applicable)

The following table provides the reader with a view of the impact to the Unilever on-premises networks for the deployment. Note. The reference (**Ref**) number is taken from the above table. **Important.** If interfaces have been entered in this section a Network Impact Assessment (NIA) will be required.

The UTC time zone is used and if needed the local time zone difference is specified as a footnote under the table.

Ref From 6.1	Peak(s) or Schedule Time(s) and Days	Firewall Rule?	Max # Per Hour	Max Size	Average Per Hour	Average Size
1	8AM – 5PM UTC / Weekdays	Y	100	100MB	80	10KB

Local User / IT Service Time Zone: IST (+ 5 Hrs 30Mins).

7. SECURITY COMPLIANCE & DATA

The following section provides the reader with a view of the security considerations taken into account for the design.

Category	Description
Physical Location Security Considerations	The environment is secured in the Azure West Europe Region and is compliant with Unilever Data Center Security Standards.
Data access & governance (Authorisation), e.g., new user request process	• All new user requests will be initiated via the Self-Service Request Tool workflows. The application owner will approve access. • Unilever R&D users will access the application in UL offices via Express Route & CXP Firewall and via ZPA when working from home. • Unilever R&D users will access the AFS from Unilever office and via ZPA when working from home. • IT Support (HCL) will connect the application server using RDP via Citrix. • UL Application Support will connect the application server using RDP via Citrix. • No Change in the existing user list while migrating from On-prem to Cloud.
Directory Services & Authentication	• The current service/admin accounts will be mapped to new servers and will be authenticated from AD. • UL R&D users and application support will access the Azure Storage BLOB via its public endpoint using SAS token. • User access to the application is managed by AD Group. Users are added to following AD Group to grant access to the application. **SEC-AZI-PipelinePilot**
Privileged access accounts and service/API user accounts	• Administrator privileges would be provided to the following users to install and configure the application with their respective admin credentials. ○ UNIL-ADMN-Ian.wood ○ UNIL-ADMN-Ian.Stott
Server security, perimeter security & changes to Unilever firewalls required?	Citrix will be used to ensure no direct user connectivity from the Internet.
Legal & Regulatory Compliance	None

Category	Description
Information Security Considerations (including safeguard of sensitive and personal data)	• Most of the TCP connections in the cloud are either HTTPS connections or are TLS v1.2 enabled connections. • There is an HTTP port open on the server, which is used to facilitate tomcat management, wherein, connection is initiated from the server loopback address – locally. • An SFTP Connection is initiated from cgtsapp24463 by the application. It's an inbuilt function in the app to fetch data/files from a location internally.
If there is a PaaS (not Unilever Azure/AWS PaaS) or SaaS provider involved in this solution, does the service provider meet targets specified in the ICAA.	Not available.
Backup platform, type of backup & scheduling for servers and data	DELL EMC Networker will be used for the application servers. The existing backup policy applies. IaaS servers are backed up using Networker once every 24 hours
Patching schedule, service hours and maintenance windows	• Standard patching schedules and planned maintenance windows apply. • Patching of the server will be the responsibility of the HCL cloud infra management team.
Data archiving, cleansing & purging (life cycle management)	Not available.
Data migration	• Accenture to build the server, provision AFS & BLOB in the new Azure cloud environment. • Application team will require internet access on the server for 5 days, to download the required software and tools and for configuration and setup of the application.
Data and system audit & monitoring	Not available.

8. USER INFORMATION

The following section provides the reader with a view of the user considerations taken into account for the design.

8.1 User Information

Category	Description
Supplier Relationships & Application Support	Infrastructure Build - Accenture Application installation and configuration – Unilever and third-party vendors (Capgemini). IT Infra Support / Maintenance – UL/HCL.
Geography & User base	Users will remain the same as they are in the current environment. The total number of R&D users (Global) is – 60
Impact to Unilever clients (Workstation / laptop client OS or applications)	None

9. LOW LEVEL DEPLOYMENT CONFIGURATION

The following section provides the reader with a detailed view of the solution deployment.

It's important to note that detailed information regarding the following can be **found in the build sheets created for this design**. Note. The list is not exhaustive.

- Server names.
- Server VM cloud types.
- Server drive configuration.
- SQL server database information.
- Cloud subnet information.
- Cloud resource tag information.
- Key Vault information.
- Cloud resource groups.
- Security Group, Network Security Group and Application Security Group information.
- Detailed backup information.
- Azure / AWS PaaS information.
- Cloud Instance On-Times (ParkMyCloud configuration detail).

9.1 Hosting Model & Location

Pipeline Pilot – R&D, Global application hosting server be deployed in Azure West Europe region (Amsterdam) service plan chosen Pay-As-You-Go.

9.2 DNS & Certificate Information

The following table provides the reader with an understanding of the DNS addresses required for the deployment. It also provides the reader with an understand of whether a certificate will be required for a Web service or an application component. Note. Does not include server DNS addresses but rather user access, API and application DNS addresses required.

DNS Address / Application Component	Role	Certificate Required?	Internal / External Certificate?
rdplp-sbx.unilever.com	Application front end access	Y	External

9.3 Internet Proxy and NSG / SG Whitelists

The following table provides the reader with an understanding whitelist requirement for the solution.

DNS Address / IP Address	Role	Internet Proxy or NSG/SG Whitelist?
-	-	-

9.4 Production (& DR) – Environments, Size & Software

The following table provides the reader with a list of components deployed for production and DR as part of this solution.

Topology Reference	Role / Software	SBB Name / Infrastructure Module	SBB Sizing	HA (Y/N)	ParkMy Cloud (Y/N)	
NA	NA	NA	NA	NA	NA	

9.5 Non-Production – Environments, Size & Software

The following table provides the reader with a list of components deployed for non-production as part of this solution.

Topology Reference	Role / Software	SBB Name / Infrastructure Module	SBB Sizing	HA (Y/N)	ParkMy Cloud (Y/N)
SB01/A	App Server	inf-azure-iaas-vm-001	**Standard_F16s_v2:** 16 vCPUs, 32GB RAM Y:\ with target from AFS Z:\ with target from AFS C:\ Standard_E6 (64G) Upto 500 IOPS SSD Disk D:\ Standard_E10 (128G) Upto 500 IOPS SSD Disk	N	Y

10. APPENDICES

The following section provides the reader with additional context for the design that does not necessarily belong in the main body of the document.

10.1 Additional Notes

10.1.1 Subscription Detail

Subscription Name	R&D-In Silico First eScience-01
Subscription ID	01491751-2545-4f64-afa7-54ffc907183d
Account Name	R&D - In Silico First (eScience)
Environment	Sandbox

10.1.2 Existing Solution Documentation

Existing SDL

10.1.3 Ambers from SDB Review

Amber	Mitigation/Comments
Topology - RDP protocol missing.	Added.
Topology - User connections to BLOB and update to show permanent.	Removed *temporary comment in Visio and added connectivity details for BLOB. Also, updated the LLD to show connectivity details and Data Access detail in security section.
Security Section (Data Access) - Provide access information for the BLOB storage.	This has been added.
Topology - Clarify SFTP client/server connection and software used/managed.	Connection is initiated from PLP server. It's an inbuilt function in PLP to fetch data/files from a location internally. No explicit SFTP client is required to be installed on the server.
Information Security - Update section as more than HTTPS and TLS is used	This has been updated.

10.1.4 Migration Run Book

The following URL provides details of the run book when migrating from On-Premises to the cloud.

Reference	Description	URL
Global Architecture Repository (GAR)	The main architecture repository for Unilever.	https://unilever.sharepoint.com/teams/GlobalArchitectureRepository/SitePages/HomePage.aspx
Draft Design Documentation	Should include infrastructure requirements, design, Visio diagram, ICAA, etc.	https://unilever.sharepoint.com/teams/GlobalArchitectureRepository/Working%20Documents/Forms/AllItems.aspx
Published Design Documentation	Contains the publish design document.	https://unilever.sharepoint.com/teams/GlobalArchitectureRepository/Published1/Forms/AllItems.aspx#InplviewHash0d5ca14d-ba5a-4565-9a7b-2e2c3373db94=SortField%3DCreated-SortDir%3DDesc
Reference Architecture	Current and destination architecture provided by the Enterprise Architecture.	https://refarch.unilever.com/refarch.php
Information (Data) Classification	Describes the different data classification levels Unilever applies to its data.	https://unilever.sharepoint.com/sites/IPZ/Standard%20Documents/Information%20Classification%20Standard.pdf
IT Service Continuity Levels	Describes the Service Criticality (SC), Disaster Recovery (DR, also known as RTO) and data Recovery Point Objective (RPO) levels applied within Unilever.	https://unilever.sharepoint.com/sites/IPZ/Standard%20Documents/Security%20Standards/IT%20Service%20Continuity.pdf
Technical Security Standards	Provides the security controls Unilever require to be in place across business and technology domains.	https://unilever.sharepoint.com/sites/IPZ/Pages/Policy-and-Standards.aspx?list=Technical%20Security%20Standards
Secure By Design (SBD) Process	The SBD process including the link to the ICAA process.	https://unilever.sharepoint.com/sites/IPZ/Pages/Secure-by-design-home.aspx
Information Security Assessment (ISA) / Self Assessment (SA)	The SA / ISA process	https://unilever.sharepoint.com/sites/IPZ/Pages/Information-Security-Assessment.aspx
Risks or Exceptions to Policy (ETP)	The Risk and ETP processes maintained by the Information Security team.	https://unilever.sharepoint.com/sites/Securevault_Information%20Security%20ETP%20%26%20Risk%20Repository/SitePages/ETP-Landing.aspx

Early Build Exception	Request a build before design board governance (needs to be approved by Advice & Guidance before submission).	https://unilever.sharepoint.com/teams/GlobalArchitectureRepository/Lists/Build%20Process%20Exception%20Tracker/NewForm.aspx
Dublin Build Exception	Request a build in the Azure Dublin (old) region (needs to be approved by Advice & Guidance before submission).	https://unilever.sharepoint.com/teams/GlobalArchitectureRepository/Lists/Azure%20Dublin%20Build%20Exception/AllItems.aspx
EA SC / DR Standards	Enterprise Architecture Service Criticality and Disaster Recovery Principles.	TBC
EA Hosting Standards	Enterprise Architecture hosting standards.	TBC
Composite SLA Information	Information regarding composite service levels.	TBC
AlgoSec Firewall Process	Process to update On-Premise firewall rules.	https://unilever.sharepoint.com/sites/IPZ/Pages/Algosec-FireFlow-Firewall-Request-Process.aspx
Akamai WAF Process	process to onboard onto the Akamai WAF.	TBC
Unilever Data Retention Policy	Details data retention required for legal purposes.	https://unilever.sharepoint.com/sites/Code%20and%20Code%20Policies/Documents/Forms/AllItems.aspx (Current PDF found here https://unilever.sharepoint.com/sites/Code%20and%20Code%20Policies/Documents/Forms/AllItems.aspx)

10.2 Design Reference Documentation

The following table provides the reader with useful URL links to help better understand the concepts applied and context for this design.